Proof notes on topology

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Contents

1 Metrical Spaces **1**

2 Topological Spaces **2**

3 Construction of topological spaces **6**

4 Compacity **8**

5 Connection **13**
5.1 Connected sets 13
5.2 Arc-connected sets 16

6 Aplications to the topology of $\mathbb{R}^2$ **18**
6.1 Definitions 18
6.2 Theorems 18

1 Metrical Spaces

Proposition 1.1. *Let (X, d) be a metrical space. Then:*

1. *If $q \in B(p, \varepsilon) \implies \exists \delta > 0$ such that $B(q, \delta) \subseteq B(p, \varepsilon)$*
2. *If $q \in \bigcap_{i=1}^{r} B(p_i, \varepsilon_i)$ for $p_i \in X$ and $i = 1, ..., r$. Then $\exists \delta > 0$ such that $B(q, \delta) \subseteq \bigcap_{i=1}^{r} B(p_i, \varepsilon_i)$*

Proof. **(1)**
If $q \in B(p, \varepsilon) \implies d(p, q) < \varepsilon$, we take $\delta = \frac{\varepsilon - d(p,q)}{2}$.
We are going to see that if $q' \in B(q, \delta) \implies q' \in B(p, \varepsilon)$.
If $q' \in B(q, \delta) \implies d(q', p) \leq d(q', q) + d(q, p) \leq \frac{\varepsilon - d(p,q)}{2} + d(p, q) \leq \frac{\varepsilon + d(p,q)}{2} \leq \varepsilon$.
(2)
We have that $q \in \bigcap_{i=1}^{r} B(p_i, \varepsilon_i)$ this implies that $q \in B(p_i, \varepsilon_i)$ $\forall p_i \in X$ for $i = 1, ..., r$.
Individually $\exists \delta_i$ such that $B(q, \delta_i) \subseteq B(p_i, \varepsilon_i) \forall i = 1, \ldots, r$.
We take $\delta = min_{i=1,\ldots,r}\{\delta_i\}$ and obviously $B(q, \delta) \subseteq \bigcap_{i=1}^{r} B(p_i, \varepsilon_i)$. ■

Definition 1.1. We say that a function $f : (X, d) \to (x', d')$ i continuous $\iff \forall p \in X$ $\forall \varepsilon > 0$ $\exists \delta > 0$ such that $f(B_d(p, \delta)) \subseteq B'_d(f(p), \varepsilon)$

Theorem 1.2. *Let be $f : (X, d) \to (x', d')$.*
Then we have

$$f \text{ is continuous } \iff \forall V \subseteq X',\ f^{-1}(V) \subseteq X \text{ is an open} \tag{1}$$

Proof. $\boxed{\Longleftarrow}$
We are taking $p \in X$ and $\varepsilon > 0$. We define $V = B'_d(f(p), \varepsilon) \subseteq X'$ an open. By the hypothesis $f^{-1}(V)$ is an open of X that has the point p. Because it is an open $\exists \delta > 0$ such that $B_d(p, \delta) \subseteq f^{-1}(V) = f^{-1}(B'_d(f(p), \varepsilon))$. Immediately we have $f(B_d(p, \delta)) \subseteq B'_d(f(p), \varepsilon)$ and then f is continuous.
$\boxed{\Longrightarrow}$

We are taking $p \in X$ and V open set of X' such that $f(p) \in V$. Since V is an open set $\exists \varepsilon > 0$ such that $B_{d'}(f(p), \varepsilon) \subseteq X'$. By hypothesis $\exists \delta > 0$ such that $f(B_d(p, \delta)) \subseteq B'_d(f(p), \varepsilon) \subseteq V$ it's an open set. Then $B_d(p, \delta) \subseteq f^{-1}(V)$ and $f^{-1}(V)$ is an open set. ■

2 Topological Spaces

Proposition 2.1. *Let (X, τ) be a Hausdorff e.t and $p \in X$. Then $\{p\}$ is a closed set.*

Proof. Simply we have to see that if $D = \{p\}$ then D^c is an open set.
Let be $q \in X \setminus \{p\}$ a point. Because of X is Hausdorff $\exists U, V_q \in \tau$ with $p \in U$ and $q \in V_q$ such that $U \cap V_q = \emptyset$. Then $V_q \subseteq X \setminus U \subseteq X \setminus \{p\}$ $(p \in U)$.
Finally $\bigcup_{q \in X \setminus \{p\}} V_q \subseteq X \setminus \{p\}$. Then $X \setminus \{p\}$ is an open set because it's the union of open sets. ■

Proposition 2.2. *Let (X, τ_1) and (X, τ_2) two topologies on the same set with the respective basis $\mathcal{B}_1, \mathcal{B}_2$.*
Then,

$$\tau_1 \leq \tau_2 \iff \forall B_1 \in \mathcal{B}_1,\ \forall p \in B_1,\ \exists B_2 \in \mathcal{B}_2 \text{ such that } p \in B_2 \subseteq B_1 \tag{2}$$

Proof. $\boxed{\Longleftarrow}$
We want to see that if $U \in \tau_1 \implies U \in \tau_2$.
We take $p \in U \in \tau_1$. Then $\exists B_1 \in \mathcal{B}_1$ such that $p \in B_1 \subseteq U$. By hypothesis, $\exists B_2 \in \mathcal{B}_2$ such that $p \in B_2 \subseteq B_1 \subseteq U$ and then $U \in \tau_2$ by definition of basis.

$\boxed{\Longrightarrow}$
Let $B_1 \in \mathcal{B}_1$ and let $p \in B_1$. Since $B_1 \in \tau_1$ and $\tau_1 \subseteq \tau_2$, we have $B_1 \in \tau_2$. Since $\mathcal{B}_2$ is a basis for τ_2 and $p \in B_1$, there exists $B_2 \in \mathcal{B}_2$ such that

$$p \in B_2 \subseteq B_1.$$

■

Definition 2.1. We define the adherence of A as:

$$\bar{A} = \bigcap_{C \text{ closed and } A \subseteq C} C$$

Definition 2.2.

$$\mathring{A} = \bigcup_{U\in\tau,\ U\subseteq A} U$$

Proposition 2.3. *Let (X,τ) be a topological space $A \subseteq X$. Then:*

1. $p \in \mathring{A} \iff \exists U \in \tau$ *such that* $p \in U \subseteq A$
2. $p \in \bar{A} \iff \forall U \in \tau$, *if* $p \in U$ *then* $U \cap A \neq \emptyset$.

Proof. The first is trivial by the definition of interior.
For the second one:
$\boxed{\Longrightarrow}$

Let be $U \subseteq \tau$ such that $p \in U$. Suppose that $U \cap A = \emptyset$. Then $A \subseteq U^c \iff \bar{A} \subseteq U^c$, then $p \notin \bar{A}$. Contradiction.
$\boxed{\Longleftarrow}$

We now have $U \subseteq \tau$ such that $p \in U$ and $U \cap A \neq \emptyset$.
Suppose $p \notin \bar{A} = \bigcap_{C \text{ closed}} C$. Then it exists C closed such that $p \notin C$, then $p \in C^c = U$ open set such that $U \cap A = \emptyset$. ■

Definition 2.3. $p \in A' \iff \forall U \in \tau \implies U \cap (A \setminus \{p\}) \neq \emptyset$

Proposition 2.4. $\bar{A} = A \cup A'$

Proof. $\boxed{\subseteq}$
If $p \in \bar{A} \implies \forall U \subseteq \tau\ U \cap A \neq \emptyset$. If $p \in A$ we are done. In the other case we have $U \cap A \setminus \{p\} \neq \emptyset$ and then $p \in A'$.
$\boxed{\supseteq}$

If $p \in A \cup A'$ we have two possible cases:

1. $p \in A$ and then $p \in \bar{A}$ by definition ($\bar{A}$ is the smallest closed set that contains A).
2. $p \in A'$ then for all $U \subseteq \tau$ we have $U \cap A \setminus \{p\} \neq \emptyset$ and immediately we have $U \cap A \neq \emptyset$ so $p \in \bar{A}$.

■

Definition 2.4. if (X,τ) is a top.s. Then (X,τ) is separable $\iff \exists A \subseteq X$ such that $\bar{A} = X$ (Dense) and it's countable.

Proposition 2.5. *Let be (X,τ) a top.s that its 2AN (that's that has a countable basis). Then (X,τ) is separable.*

Proof. If (X,τ) is a 2AN topological space then it exists $\mathcal{B}$ a countable basis with $B_n \subseteq \mathcal{B} \forall n \in \mathbb{N}$. $\forall x \in \mathbb{N}$ we define $a_n \in B_n \subseteq \mathcal{B}$. Then we define $A = \{a_n\}_{n\in\mathbb{N}}$ the subset A that we are going to see if it's dense and countable.

1. A is countable by construction.

2. Let's see that A is dense, that's $\bar{A} = X$.
   Let's take $p \in X$ and $U \subseteq \tau$ such that $p \in U$. Then $\exists B_n \in \mathcal{B}$ such that $p \in B_n \subseteq U$ by def. of basis. Because of $a_n \in U$ then $U \cap A \neq \emptyset$ and $p \in \bar{A}$

■

The reciprocal is true if (X, d) is a metrical space.

Proposition 2.6. *If (X, d) is a metrical space such that $\tau = \tau_d$, then (X, τ_d) is separable $\implies$ (X, τ) is 2AN.*

Proof. If (X, τ_d) is separable it exists $A \subseteq X$ such that $\bar{A} = X$ and A is countable.
We define the following base:

$$\mathcal{B} = \{B_d = B(a, \frac{1}{n}) | a \in A, n \in \mathbb{N}\} \tag{3}$$

This is like $(\mathbb{N} \times \mathbb{N})$. We are going to see if this is a basis.
Let be $U \subseteq \tau$ such that $x \in U$.

1. We are going to take $n \in \mathbb{N}$ such that $B_d(x, 1/n) \subseteq U$.

2. Because A is dense, it $\exists a \in A$ such that $a \in B_d(x, \frac{1}{2n})$

To proof that $\mathcal{B}$ is a base, we have to see that $x \in B_d(a, 1/n) \subseteq U$.
The first part is trivial ($x \in B_d(a, 1/n) \iff a \in B_d(x, 1/n)$).
For the second part we are taking $y \in B_d(a, \frac{1}{2n})$ and we are seeing that $y \in U$. Obviously, $d(x, y) \leq d(x, a) + d(y, a) \leq \frac{1}{2n} + \frac{1}{2n} = \frac{1}{n}$. And then $y \in B_d(x, \frac{1}{n}) \subseteq U$.
Then $x \in B_d(a, \frac{1}{n}) \subseteq U$ and $\mathcal{B}$ is a base and (X, τ) is 2AN. ■

Definition 2.5. A subbase $\mathcal{S}$ defines a base in the following way: $\mathcal{B} = \{\bigcap_{i \in I} S_i | S_i \in \mathcal{S}\}$ (the set of the unions of all the arbitrary intersections of elements of the subbase).

Proposition 2.7. *If we have $(X, \tau_X) \xrightarrow{f} (Y, \tau_Y)$ and $\mathcal{S}, \mathcal{B} \subseteq \tau_Y$ subbase and base respectively of τ_Y then:*

1. *f cont $\iff$ $\forall B \in \mathcal{B}$, $f^{-1}(B) \subseteq \tau_X$.*

2. *f cont $\iff$ $\forall S \in \mathcal{S}$, $f^{-1}(S) \subseteq \tau_X$.*

Proof. $\boxed{\implies}$ It's direct by definition in the two cases.
$\boxed{\impliedby}$

Let be $V \subseteq \tau_Y$, then $V = \bigcup_{i \in I} B_i$ such that $B_i \in \mathcal{B}$.
Then $f^{-1}(V) = f^{-1}(\bigcup_{i \in I} B_i) = \bigcup_{i \in I} f^{-1}(B_i) \subseteq \tau_X$. ■

Proposition 2.8. *Let be $(X, \tau_X) \xrightarrow{f} (Y, \tau_Y)$. There are equivalent:*

1. *f cont.*

2. *$\forall A \subseteq X$, $f(\bar{A}) \subseteq \overline{f(A)}$*

3. *$\forall C$ closed set, $f^{-1}(C)$ is closed set.*

Proof. (1) $\implies$ (2)
Let be $A \subseteq X$ and $x \in \bar{A}$. Then obviously $f(x) \in f(\bar{A})$. We want to see $f(x) \in \overline{f(A)}$.
Let $V \subseteq \tau_Y$ an open set such that $f(x) \in V$. Then $f^{-1}(V)$ is an open set by hypothesis. Of course, because of $x \in f^{-1}(V)$ and $x \in \bar{A}$, we have $f^{-1}(V) \cap A \neq \emptyset$.
We are taking $a \in f^{-1}(V) \cap A$ and then $f(a) \in V \cap f(A) \implies V \cap f(A) \neq \emptyset$, because $f(x) \in V \implies f(x) \in \overline{f(A)}$.
(2) $\implies$ (3)
If C is a closed set we want to see that $D = f^{-1}(C)$ is a closed set.
We have that D is closed $\iff \bar{D} = D \iff \bar{D} \subseteq D$.
By hypothesis $f(\bar{D}) \subseteq \overline{f(D)} = \overline{f(f^{-1}(C))} \subseteq \bar{C} = C$(C is closed). Then $\bar{D} \subseteq f^{-1}(C) = D$ and D is closed.
(3) $\implies$ (1)
If we have $V \subseteq \tau_Y$ closed set we want to see $f^{-1}(V) \subseteq \tau_X$. Let be $V = C^c$ where C is a closed set. Then $f^{-1}(C^c) = (f^{-1}(C))^c \subseteq \tau_X$. ■

Definition 2.6. $\{x_n\}_{n\in\mathbb{N}} \to x \iff \forall U \in \tau_x$ such that $x \in U \exists n_0$ such that $\forall n \geq n_0\ x_n \subset U$

Proposition 2.9. *Let be (X, τ) a topological Hausdorff space and $\{x_n\}_{n\in\mathbb{N}} \subset X$ a sequence. Then if $\{x_n\}_{n\in\mathbb{N}}$ has a limit, this one is unique.*

Proof. Supose that $\{x_n\}$ has two limits l_1, l_2. Then there exists $U, V \in \tau_X$ such that $U \cap V = \emptyset$.
By def. of limit $\exists n_1$ such that $\forall n \geq n_1,\ x_n \subset U$.
By the same reasoning $\exists n_2$ such that $\forall n \geq n_2,\ x_n \subset V$.
If we take $n_0 = max\{n_1, n_2\}$ then $x_n \in U \cap V$!!! ■

Proposition 2.10. *Let (X, τ) be a Hausdorff 2AN topological space. Then:*

1. *If $A \subseteq X$, $p \in \bar{A} \iff \exists \{x_n\}_{n\in\mathbb{N}} \subset A$ such that $\lim x_n = p$*
2. *If $A \subseteq X$, $p \in A' \iff \exists \{x_n\}_{n\in\mathbb{N}} \subset A \setminus \{p\}$ such that $\lim x_n = p$*
3. *f is continuous $\iff \forall a \in X\ \forall \{x_n\} \subseteq X$, $a = \lim x_n \implies f(a) = \lim f(x_n)$*

Proof. **(1)**
$\boxed{\Longleftarrow}$

If we have $\{x_n\} \subseteq A$ such that $\lim x_n = p$ it means that $\exists U \in \tau_X$ and $n_0 \in \mathbb{N}$ such that $p \in U$ and $\forall n \geq n_0\ x_n \subseteq U$. Then $U \cap A \neq \emptyset$ and $p \in \bar{A}$.
$\boxed{\Longrightarrow}$

We have $p \in \bar{A}$ and we define the base (countable):

$$\mathcal{B}_p = \{U \in \tau_X | p \in U\} = \{B_n\}_{n\in\mathbb{N}} \text{ (if the set is finite, we can repeat elements)}$$

We can now build $\{x_n\}$.

1. $p \in B_1 \xrightarrow{p\in\bar{A}} A \cap B_1 \neq \emptyset \implies x_1 \in A \cap B_1$
2. $p \in B_1 \cap B_2 \xrightarrow{p\in\bar{A}} A \cap B_1 \cap B_2 \neq \emptyset \implies x_2 \in A \cap B_1 \cap B_2$

3. ...

4. $p \in B_1 \cap B_2 \cap \cdots \cap B_n \xrightarrow{p \in \bar{A}} A \cap B_1 \cap B_2 \cap \cdots \cap B_n \neq \emptyset \implies x_n \in B_1 \cap B_2 \cap \cdots \cap B_n$

We have defined $\{x_n\}$.
If we take $V \in \tau_X$ such that $p \in V$, by the def of basis $\exists B_{n_0} \in \mathcal{B}_p$ such that $p \in B_{n_0} \subseteq V$.
$\forall n \geq n_0$ we have $x_n \in A \cap B_1 \cap B_2 \cap \cdots \cap B_n \subseteq A \cap B_1 \cap B_2 \cap \cdots \cap B_{n_0} \subseteq A \cap B_1 \cap B_2 \cap \cdots \cap V \subseteq V$.
Then $\{x_n\} \to p$.
(2)
This is exactly the same as the (1) proof.
(3)
$\boxed{\implies}$

We have f continuous and a succession $\{x_n\} \to x$, we want to see if $f(\{x_n\}) \to f(x)$. It $\exists U \in \tau_Y$ such that $f(x) \in U$, because of the continuity of f, $f^{-1}(U) \in \tau_X$ and $\exists n_0 \in \mathbb{N}$ such that $\forall n \geq n_0$ $x_n \in f^{-1}(U)$ (because $x \in f^{-1}(U)$). Then $\forall n \geq n_0$ $f(x_n) \in U$.
$\boxed{\impliedby}$

We have $\forall \{x_n\} \to x \implies f(\{x_n\}) \to f(x)$. We are going to proof that $f(\bar{A}) \subseteq \overline{f(A)}$ $\forall A \subseteq X$.
We are taking $x \in \bar{A}$ and immediately we get $\{x_n\} \subset A \to x$ (by (1)). Obviously, $f(x) \in f(\bar{A})$ and by hypothesis $f(\{x_n\}) \to f(x)$, then $f(x) \in \overline{f(A)}$ because it's the limit of a succession.
Then $f(\bar{A}) \subseteq \overline{f(A)}$ $\forall A \subseteq X$ and f is continuous. ■

3 Construction of topological spaces

Theorem 3.1. ***Universal property of the subspace topology***
Let $Y \subseteq X$ and $i : Y \hookrightarrow X$ the inclusion.
Then $f : W \to Y$ is continuous $\iff$ $i \circ f : W \to X$ is continuous. (Restricting functions into subspaces are continuous if they are in the "general" space).

Theorem 3.2. *Let (X, τ_X) a topological space. $A \underset{i}{\subseteq} X$.*

1. *$\tau_{X|A}$ is the les thin topology (menys fina) such that i is continuous.*

2. *$f : W \to A$ is continuous $\iff$ $i \circ f : W \to X$ is continuous.*

Proof. **(1)**
Direct by construcion.
(2)
$\boxed{\implies}$

$i \circ f$ is continuous because it is the composition of two continuous functions.
$\boxed{\impliedby}$

If we have $U \in \tau_{X|A}$ we want to see that $f^{-1}(U)$ is an open set.
We know that $\exists V \in \tau_X$ such that $U = i^{-1}(V \cap X)$ (By def of subspace topology). Then $f^{-1}(U) = f^{-1}(i^{-1}(V \cap X)) = (i \circ f)^{-1}(V \cap X) \in \tau_X$ that's an open because $i \circ f$ is continuous. ■

Proposition 3.3. ***"Lemma d'enganxament"***
$(X, \tau_X), (Y, \tau_Y)$ topological spaces and $X = C_1 \cup \cdots \cup C_r$ where C_i is a closed set.
Then we define $f_i : C_i \to Y$ such that $f_{i_{|C_i \cap C_j}} = f_{j_{|C_j \cap C_i}}$ $\forall i, j$.
Then $f : X \xrightarrow{Y}$ defined by the f_i is continuous.

Proof. We are going to see that if D is a closed set from Y then $f^{-1}(D)$ is a closed set.
$f^{-1}(D) = \{x \in X | f(x) \in D\} = \bigcup_{i=1}^{r} \{x \in C_i | f_i(x) \in D\} = \bigcup_{i=1}^{r} f_i^{-1}(D) = \bigcup_{i=1}^{r} T_i \cap C_i =$ *closed.*
Where in the last equalty we've used that $f_i^{-1}(D) = C_i \cap T_i$ where T_i is a "random" closed set from X. Then the union of closed spaces is closed. ■

Definition 3.1. If we have an arbitrary colection of topological spaces $\{(X_i, \tau_i)\}_{i \in I}$ and we consider:

$$\prod_{i \in I} X_i \xrightarrow{\pi_i} X_i$$
$$(a_i)_{i \in I} \mapsto a_i$$

Then a subbase is:

$$\mathcal{S} = \{\pi_i^{-1}(U_i) | U_i \in \tau_i, i \in I\} = \{U_i \times \prod_{i \neq j} X_j | U_i \in \tau_i, i \in I\}$$

And a base:

$$\mathcal{B} = \{\prod_{i \in I} U_i | U_i \in \tau_i, i \in I\}$$

Theorem 3.4. *Let $\{(X_i, \tau_i)\}_{i \in I}$ be a topological space, then:*

1. *$\prod_{i \in I} \tau_i$ is the less thin topology that makes the projections $\pi_i : \prod_{i \in I} X_i \xrightarrow{\pi_i} X_i$ continuous.*

2. ***Universal property of the product topology***
   If $f_i = \pi \circ f$ then f_i is continuous $\iff$ $\pi_i \circ f$ is continuous. (That's f is continuous if it is component by component).

Proof. $\boxed{(1)}$
By construction.
$\boxed{(2)}$

$\boxed{\Longrightarrow}$

Direct by composition of functions.

$\boxed{\Longleftarrow}$

We want to see if for $U \in \prod_{i \in I} \tau_i$ then $f^{-1}(U) \in \tau_Z$.
If $U \in \prod_{i \in I} \tau_i$ then I can rewrite U in terms of the subbase of $\mathcal{S}$. $U = \bigcup S_{i_1} \cap \cdots \cap S_{i_r} = \bigcup \pi_{i_1}^{-1}(U_{i_1}) \cap \cdots \cap \pi_{i_r}^{-1}(U_{i_r})$ where S_{i_j} are elements of $\mathcal{S}$.
Then $f^{-1}(U) = f^{-1}(\bigcup \pi_{i_1}^{-1}(U_{i_1}) \cap \cdots \cap \pi_{i_r}^{-1}(U_{i_r}) = \bigcup \bigcap_{j=1}^{r} f^{-1}(\pi_{i_j}^{-1}(U_{i_j})) = \bigcup \bigcap_{j=1}^{r} (\pi_{i_j} \circ f)^{-1}(U_{i_j})) \in \tau_Z$ by hypothesis. ■

Definition 3.2. (X, τ_X) a topological space, $f : X \to Y$ an exhaustive function. $\tau_f = \{V \subseteq Y | f^{-1}(V) \in \tau_X\}$

Theorem 3.5. *1. τ_f is the less thin topology that makes f continuous.*

2. ***Universal property of the quocient topology***
If $g : (X, \tau_X) \to (Z, \tau_Z)$, $f : (X, \tau_X) \to (Y, \tau_f)$ and $h : (Y, \tau_f) \to (Z, \tau_Z)$. Then $g = h \circ f$ is continuous $\iff$ h is continuous.

Proof. $\boxed{\Longleftarrow}$

Direct

$\boxed{\Longrightarrow}$

Taking $U \in \tau_Z$ we want to see $h^{-1}(U) \in \tau_f$.
$h^{-1}(U) \in \tau_f \iff f^{-1}(h^{-1}(U)) \in \tau_X \iff (h \circ f)^{-1}(U) \in \tau_X$ which is trivial by hypothesis. ■

Proposition 3.6. ***Tallar i enganxar***
If we have $(X, \tau_X), (Y, \tau_Y)$ topological spaces such that $f : X \to Y$ is an homeomorphism. We have two equivalence clases $\underset{X}{\sim}$, $\underset{Y}{\sim}$ such that $x \underset{X}{\sim} x' \iff f(x) \underset{Y}{\sim} f(x')$, then $\exists \overline{f} : X/\sim \to Y/\sim$ homeomorphism.

Proof. We have the following commutative diagram.

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle \pi_2} \\ X/\sim & \xrightarrow[\quad \overline{f} \quad]{} & Y/\sim \end{array}$$

We define $g : X \to Y/\sim$ such as $g = \pi_2 \circ f$. g is continuous because it's composition of two continuous functions.
Obviously, $g(p) = g(p') \iff [f(p)]_Y = [f(p')]_Y \iff p \underset{X}{\sim} p'$.
Then g induces a function $h : X/\sim \to Y/\sim$. h is continuous by the universal property of the quocient topology (g is continuous).
By the same argument we can find a function $l : Y \to X/\sim$ that induces $m : Y/\sim \to X/\sim$ that again is a continuous function.
Finally we have to see that:

$$m \circ h = Id_{X/\sim}$$
$$h \circ m = Id_{Y/\sim}$$

We are only proving the first one, $(m \circ h)([p]_X) = m(h([p])_X) = m([f(p)]_Y) = [f^{-1}(f(p))]_X = [p]_X$. ■

4 Compacity

Definition 4.1. if (X, τ_X) a topological space, we say that its compact if $\forall U \in \tau_X$ and a collection $\{V_i\}_{i \in I} \in \tau_X$ an open cover of U, then $\exists i_1, \ldots, i_r \in I$ such that $U = V_{i_1} \cup \cdots \cup V_{i_r}$.

Proposition 4.1. ***Cantor***
let (X, τ_X) a compact topological space.
If $\{C_n\}_{n\in\mathbb{N}}$ collection of non-empty nested closed sets ($C_n \supseteq C_{n+1}$), then $\bigcap_{n\in\mathbb{N}} C_n \neq \emptyset$.

Proof. We are going to suppose that $\bigcap_{n\in\mathbb{N}} C_n = \emptyset$. Because of X is compact, $\exists n_1, \ldots, n_k \in \mathbb{N}$ such that $C_{n_1} \cap \cdots \cap C_{n_k} = \emptyset$ and then $C_m = \emptyset$ taking $m = max\{n_1, \ldots, n_k\}$. ■

Theorem 4.2. *If $(X, \tau_X) \xrightarrow{f} (X, \tau_Y)$ is continuous and X is compact then $f(X)$ is compact.*

Proof. We take $\{V_i\}_{i\in I} \in \tau_Y$ such that $f(X) \subseteq \bigcup_{i\in I} V_i$. Because of the continuity of f, $X \subseteq f^{-1}(\bigcup_{i\in I} V_i) = \bigcup_{i\in I} f^{-1}(V_i) \in \tau_X$ it's an open cover of X.
Because of the compacity of X, $\exists i_1, \ldots, i_k \in I$ such that $X \subseteq f^{-1}(V_{i_1}) \cup \cdots \cup f^{-1}(V_{i_k})$. Then $f(X) \subseteq V_{i_1} \cup \cdots \cup V_{i_k}$. ■

Theorem 4.3. *Let (X, τ_X) a topological compact space and $C \subseteq X$ a closed set. Then $(C, \tau_{X|C})$ is compact.*

Proof. We take an open cover of C $\{U_i\}_{i\in I} \in \tau_X$. Then because C^c is an open set, $X = C \cup C^c = \bigcup_{i\in I}(U_i) \cup C^c$. Because X is compact, $\exists i_1, \ldots, i_k \in I$ such that $X = U_{i_1} \cup \cdots \cup U_{i_k} \cup C^c$. We know that $C^c \cap C = \emptyset$, then $C \subseteq U_{i_1} \cup \cdots \cup U_{i_k}$ and C is compact. ■

Theorem 4.4. *Let (Y, τ_Y) be a topological Hausdorff space such that $C \subseteq Y$ it's a compact set. Then C is closed.*

Proof. We are going to see that C^c is an open set, that's $\forall x_0 \in C^c$, $\exists U \in \tau_Y$ with $x_0 \in U$ such that $U \subseteq C^c$.
For every point y of C we can take two open sets $U_y, V_y \in \tau_Y$ such that $x_0 \in U_y$ and $y \in V_y$ and $U_y \cap V_y = \emptyset$ (by hypothesis). Then:

$$\bigcup_{y\in C} V_y \supseteq C$$

Because of the compacity of C $\exists y_1, \ldots, y_k \in C$ such that:

$$V = V_{y_1} \cup \cdots \cup V_{y_k} \supseteq C$$

If we define $U = U_{y_1} \cap \cdots \cap U_{y_k} \in \tau_Y$ (note that $x_0 \in U$) we can see that $U \cap V = \emptyset$ because if $z \in V$ $\exists y_j, j \in \{1, \ldots, k\}$ such that $z \in V_{y_j}$ and then $z \notin U_{y_j}$ and $z \notin U$.
We have found $U \subseteq C^c$ and then C is a closed set. ■

From this three theorems we have 4 interesting corollarys.

Corollary 4.5. *If (X, τ_X) is a topological space and $\sim$ an equivalence relationship then X is compact $\implies$ $X/\sim$ is compact.*

Proof. The quotient map π is continuous, then by T1 $X/\sim$ is compact. ■

Corollary 4.6. *If $(X, \sim) \cong (Y, \sim)$ then X is compact $\iff$ Y compact.*

Proof. It exist an homeomorphism between the two quotient spaces and then exists an homeo $f : X \to Y$. This homeo f is continuous and f^{-1} is continuous, then we get the result. ■

Corollary 4.7. *If $X \xrightarrow{f} Y$ is continuous, X is compact and Y is hausdorff then f is a closed function (sends closed sets to closed sets).*

Proof. Let $C \subseteq X$ a closed set. Because of X is compact, C is compact (T2), then $f(C)$ is a compact (T1) inside a hausdorff space Y, by T3 $f(C)$ is closed. ■

Remark 4.8. If f is continuous and bijective the corollary allows us the following relation:

$$f \text{ homeo} \iff f^{-1} \text{ continuous} \iff f \text{ open} \iff f\text{closed}$$

Theorem 4.9. ***Maximum and minimum value theorem***
Let $(X, \tau_X) \xrightarrow{f} (Y, \tau_Y)$ a continuous function such that X is compact and τ_y is induced by a total order.
Then f has a maximum and minimum value.

Proof. Suppose that f doesn't have a maximum value (the procedure is analogue for the minimum value). $f(X) \subseteq \bigcup_{z \in X}(-\infty, f(z))$. Because X is compact, $f(X)$ is compact and then $\exists z_1, \ldots, z_k \in X$ such that $f(X) \subseteq \bigcup_{i=1}^{k}(-\infty, f(z_i)) \overset{f(z^*)=max\{f(z_1),\ldots,f(z_k)\}}{=} (-\infty, f(z^*))$. Because f has not a maximum $f(z^*) \notin f(X)$, nevertheless we builded the cover set by using $z \in X$ then we have a contradiction. ■

Theorem 4.10. ***Tube's lemma***
Let (X, τ_X), (Y, τ_Y) two topological spaces, Y compact, $a \in X$ and $U \in \tau_X \times \tau_Y$, $U \supseteq \{a\} \times Y$. Then $\exists W \in \tau_X$ such that $a \in W$ and $W \times Y \subseteq U$.

Proof. Take $U \in \tau_X \times \tau_Y$ such that $\{a\} \times Y \subseteq U$.
A base for $\tau_X \times \tau_Y$ is: $B = \{W \times V | W \in \tau_X, V \in \tau_Y\}$. Then for every $(a, t) = p \in \{a\} \times Y$ where $t \in Y$, $\exists W_p \in \tau_X, V_p \in \tau_Y$ such that $p \in W_p \times V_p \subseteq U$.
We can cover $\{a\} \times Y$ with open sets with the following way:

$$\{a\} \times Y = \bigcup_{p \in \{a\} \times Y} W_p \times V_p$$

Because $\{a\} \times Y$ is a compact, $\exists p_1, p_2, \ldots, p_k \in \{a\} \times Y$ such that:

$$\{a\} \times Y = \bigcup_{i=1}^{k} W_{p_i} \times V_{p_i}$$

Taking $W = \bigcap_{i=1}^{k} W_{p_i}$, that's an open set because it's a finite intersection of open sets, we have $a \in W$ (because $p \in W_p \times V_p, \forall p \in \{a\} \times Y$). Then, $W \times Y \subseteq U$. ■

Theorem 4.11. *Let $(X_1, \tau_1), \ldots, (X_n, \tau_n)$ topological spaces.*
Then, $X = (X_1 \times \cdots \times X_N, \tau_1 \times \cdots \times \tau_N)$ is compact $\iff$ $\forall i$ $(X_i, \tau_i) arecompact$

Proof. $\boxed{\implies}$
Because $\forall i$ π_i the projection is continuous and X_i is a compact, $\pi_i(X) = X_i$ is a compact set.
$\boxed{\impliedby}$

We proceed by induction over n, that's only because $X_1 \times \cdots \times X_N = (X_1 \times \ldots X_{N-1}) \times X_N)$. So let's prove it for $n = 2$. Define $X = X_1 \times X_2 = \bigcup_{i \in I} U_i$ an open cover of X. For each $a \in X_1$, $\{a\} \times X_2 \subseteq X_1 \times X_2 \subseteq \bigcup_{i \in I} U_i$ is a compact set. Then $\exists i_1, \ldots, i_k \in I$ such that $\{a\} \times X_2 \subseteq U_{i_1} \cup \cdots \cup U_{i_k} = U_a$. By the Tube's lemma, $\exists W_a \in \tau_{X_1}$ such that $\{a\} \times X_2 \subseteq W_a \times X_2 \subseteq U_a$.
Obviously, $X_1 \subseteq \bigcup_{a \in X_1} W_a$, and because the compacity of X_1, $\exists a_1, \ldots, a_k$ such $X_1 \subseteq \bigcup_{i=1}^{s} W_{a_i}$.
Finally, $X1 \times X_2 = \pi_i^{-1}(W_{a_1}) \cup \cdots \cup \pi_i^{-1}(W_{a_s}) = (W_{a_1} \times X_2) \cup \cdots \cup (W_{a_s} \times X_2) \subseteq U_{a_1} \cup \cdots \cup U_{a_s}$, that is a finite union of covers of the inicial cover. ■

Theorem 4.12. ***Tychonoff's theorem***
Let $\{(X_i, \tau_i)\}_{i \in I}$ a topological space, $X = \prod X_i$ and $\tau_X = \prod \tau_i$.
Then X is a compact set $\iff$ X_i are compact sets $\forall i \in I$.

Proof. $\boxed{\implies}$
Because $\forall i$ π_i the projection is continuous and X_i is a compact, $\pi_i(X) = X_i$ is a compact set.
$\boxed{\impliedby}$

Let's suppose that X is not a compact set. We define:

$$\mathcal{S} = \bigcup_{i \in I} S_i$$

a subbase of the topology τ_X, where $S_i = \pi_i^{-1}(\tau_i)$.
By Alexander's lemma and because X is not compact $\exists$ a cover of X made of elements of $\mathcal{S}$ that has not a finite subcover. This is equivalent to $X = \bigcup_{j\in J} S_j$ where $S_j \in \mathcal{S}$.
We define $J_i = \{j \in J | S_j = \pi_i^{-1}(\tau_i)\}$ ($j \in J_i$ if $S_j = \pi_i^{-1}(U_j)$, $U_j \in \tau_i$).
Then we can rewrite:

$$X = \bigcup_{i\in I} \bigcup_{j\in J_i} S_j$$

We state that $\boxed{X_i \neq \bigcup_{j\in J_i} U_j}$. (if not, $X_i = \bigcup_{j\in J_i} U_j$, and because X_i is a compact set, $\exists j_1, \ldots, j_k \in J_i$ such that $X_i = \bigcup_{i=1}^{k} U_{j_i}$ and $X = \pi_i^{-1}(X_i) = S_{j_1} \cup \cdots \cup S_{j_k}$!! because X has not a finite subcover).
Then $\forall i \in I$ $\exists x_i$ such that $x_i \notin \bigcup_{j\in J_i} U_j$. Taking $x = (x_i)_{i\in I} \in X$ such that $\forall i \in I$ $x_i \notin \bigcup_{j\in J_i} U_j$ we have that $x \in \pi_i^{-1}(x_i) \notin \bigcup_{j\in J_i} S_j$ $\forall i \in I$ what implies $x \notin \bigcup_{i\in I} \bigcup_{j\in J_i} S_j = X$ what it's a contradiction. ■

Remark 4.13. If $C \subseteq X$ where C is a compact set and X a Hausdorff set and we have $p \notin C$, then $\exists U \supseteq C$, $U \in \tau_X$ and $p \in V \in \tau_X$ such that $U \cap V = \emptyset$.

Proposition 4.14. *Let (X, τ) be a Hausdorff compact topological space. Then X is a Normal space ($\forall C, D \subseteq X$ closed sets ($C \cap D = \emptyset$) $\exists U, V \in \tau$ such that $U \supseteq C$, $V \supseteq D$ and $U \cap V = \emptyset$)*

Proof. Let $C, D \subseteq X$ be two closed spaces. Because they are closed inside a Hausdorff space, they are compact.
Let $p \in C$, then $\exists U_p, V_p \in \tau$ such that $U_p \supseteq D$ and $p \in V_p$.
Obviously, $C \subseteq \bigcup_{p\in C} V_p$, and because C is compact $\exists p_1, \ldots, p_k \in C$ such that $C \subseteq V_{p_1} \cup \cdots \cup V_{p_k} = V$. We define $U = U_{p_1} \cap \cdots \cap U_{p_k} \supseteq D$. Then $U \cap V = \emptyset$ (if $q \in U \cap V \implies \exists j \in \{1, \ldots, k\}$ such that $q \in V_{p_j}$, but $q \notin U_{p_j}$). ■

Lemma 4.15. *(uyasho) Let (X, τ_X) a normal topological space. Let $C, D \subseteq X$ two closed sets such that $C \cap D = \emptyset$.*
Then $\exists f : X \to [0, 1]$ continuous such that $f(C) = 0$ and $f(D) = 1$.

Theorem 4.16. ***Metrizability theorem***
Let (X, τ_X) a topological space. Then, X a normal set + 2AN $\implies$ X it's metrizable ($\exists d$ distance such that $\tau_X = \tau_d$).

Proof. Define $J = \{f : X \to [0, 1 | f \text{ is continuous}]\}$ and $F : X \to F(X)$ such that $F(x) = (f_j(x))_{j\in J}$. We are going to prove that $X \cong F(X) \subseteq \mathbb{R}^J$.

1. F is an exhaustive map by definition.

2. if $p, q \in X$ such that $p \neq q$ then by uyasho's lemma $\exists f : X \to [0, 1]$ continuous map such that $f(p) = 0$ and $f(q) = 1$, hence $F(p) \neq F(q)$.

3. $F : X \to F(X)$ is continuous $\iff$ $F : X \to \mathbb{R}^J$ is continuous $\iff$ $\forall j \in J, f_j$ is continuous, what's right.

Now we are going to prove that F is an open function and hence defines an homeomorphism between X and $F(x)$.
Let $U \in \tau_X$, we want to see that if $z \in F(U) \implies z \in int(F(U))$.
If $z \in F(U) \implies \exists x \in U$ such that $F(x) = z$. Obviously $\{x\}, X - U$ are two closed sets and by uyasho's lemma $\exists f_0 : X \to [0,1]$ such that $f(X-U) = 0$ and $f_0(\{x\}) = 1$. Taking $W = \pi_{f_0}^{-1}((0,\infty))$ that's an open from $\mathbb{R}^J$ where $\pi_{f_0}(F(X)) = f_0(X)$.
Note that $z \in W \cap F(X)$ because $z = F(x)$ in the component f_0 is equal to 1.
Now we are going to see that if $z' \in W \cap F(X)$ then $\exists x' \in X$ such that $F(x') = z'$. If $z' \in W \implies f_0(x') > 0 \implies x' \notin X - U \implies x' \in U \implies F(x') \in F(U)$. Then we're done because z is in the interior of $F(U)$ and hence it's an open.

Finally we would have to use the $2AN$ property to see that $X \cong F(X) \subseteq \mathbb{R}^{\mathbb{N}}$ that's a metrical space. ■

5 Connection

5.1 Connected sets

Definition 5.1. X is not a connected set $\iff \exists U, V \in \tau_X$ such that:

1. $U \neq \emptyset$
2. $V \neq \emptyset$
3. $U \cap V = \emptyset$
4. $X = U \cup V$

Alternatively we can define connection for $X \subseteq (Y, \tau_Y)$ wjere $\tau_X = \tau_{Y|X}$. We say that X is no a connected set $\iff \exists U, V \in \tau_X$ such that:

1. $X \cap U \neq \emptyset$
2. $X \cap V \neq \emptyset$
3. $X \cap U \cap V = \emptyset$
4. $X \subseteq U \cup V$

Proposition 5.1. *If $X \subseteq (\mathbb{R}, \tau_{euc})$, then X is a connected set $\iff$ X is an interval ($\forall a, b \in X$ if $a \leq c \leq b$ then $c \in X$).*

Proof. $\boxed{\implies}$
If X is not an interval, $\exists a, b \in X$ and $\exists c$ such that $a \leq c \leq b$ and $c \notin X$. Defining $U = (-\infty, c)$ and $V = (c, \infty)$ we have:

1. $X \cap U \neq \emptyset$
2. $X \cap V \neq \emptyset$

3. $X \cap V \cap U = \emptyset$
4. $X \subseteq U \cup V$

$\boxed{\Longleftarrow}$

We have that X is connected. Then if we have $U, V \in \tau_X$ such that:

1. $X \cap U \neq \emptyset$
2. $X \cap V \neq \emptyset$
3. $X \subseteq U \cup V$

We are going to see that $X \cap U \cap V \neq \emptyset$. Taking $a \in U$ and $b \in V$ and supposing without lose of generality $a < b$ we define the set:

$$C = \{c \in X | c \in U \text{ and } c < b\}$$

Because $C \neq \emptyset$ ($a \in C$) and it's bounded, $\exists c = \sup C$ and $a \leq c \leq b$. Because X is an interval, $c \in X$.
Because of the (3) property, $X \subseteq U \cup V$ and then $X = (U \cap X) \cup (V \cap X)$.
Supposing $X \cap U \cap V = \emptyset$ we get $(X \cap V)^c = X \cap U$ and $(X \cap U)^c = X \cap V$. FInnaly;

1. Because $c = \sup C$, $c \in \overline{U \cap X}$, but $U \cap X$ is a closed set, then $c \in U \cap X$.
2. Because X is an interval and $\forall d > c$, $d \in V \implies (c, b] \subseteq V \cap X$ ($c \in X$), this implies $c \in \overline{(c, b]}$ and because $V \cap X$ is closed, $c \in V \cap X$

Then $c \in V \cap U \cap X$ what it's a contradiction and X is connected.

■

Theorem 5.2. *if $(X, \tau_X) \xrightarrow{f} (Y, \tau_Y)$ is a continuous map and X is a connected set then $f(X)$ is a connected set.*

Proof. Suppose that $f(X)$ is not a connected set. Then $\exists U, V \in \tau_Y$ such that:

1. $U \cap f(X) \neq \emptyset \implies f^{-1}(U) \cap X \neq \emptyset$
2. $V \cap f(X) \neq \emptyset \implies f^{-1}(V) \cap X \neq \emptyset$
3. $U \cap V \cap f(X) = \emptyset \implies f^{-1}(U) \cap f^{-1}(V) \cap X = \emptyset$
4. $f(X) \subseteq U \cup V \implies X \subseteq f^{-1}(U) \cup f^{-1}(V)$

Because f is a continuous map then $f^{-1}(U), f^{-1}(V) \in \tau_X$ and then X is not a connected set ■

Corollary 5.3. *X connected set $\implies X/\sim$ connected set.*

Proof. The quotient map is continuous and X is a connected set. ■

Corollary 5.4. *$X \cong Y \implies X$ connected $\iff Y$ connected*

Proof. Direct by definition ■

Corollary 5.5. ***TVI***
$(X, \tau_X) \xrightarrow{f} (\mathbb{R}, \tau_{eucl})$ a continuous map.
If X is a connected set and $x_1, x_2 \in X$ (supposing $f(x_1) \leq f(x_2)$), then $\forall x \in \mathbb{R}$ such that $f(x_1) \leq c \leq f(x_2) \exists x \in X$ such that $f(x) = c$.

Proof. Because X is a connected set and f is a continuous map, $f(X) \subseteq (\mathbb{R}, \tau_{eucl})$ is a connected set. f must be an interval (Prop 5.1). Then if we have $c \in \mathbb{R}$ *and* $f(x_1) \leq c \leq f(x_2)$ $(f(x_1), f(x_2) \in f(X))$ then $c \in f(X)$ and $\exists x' \in X$ such that $f(x') = c$. ■

Corollary 5.6. ***Bolzano's theorem***
$[a, b] \xrightarrow{f} \mathbb{R}$ a continuous map and $f(a) \cdot f(b) \leq 0$ then $\exists x_0 \in [a, b]$ such that $f(x_0) = 0$

Proof. TVI conditions. ■

Corollary 5.7. ***Fixed point theorem***
$[0, 1] \xrightarrow{f} [0, 1]$ a continuous map, then $\exists x \in [0, 1]$ where $f(x) = x$.

Proof. Define $g(x) = f(x) - x$, then we note that $g(0) = f(0) \geq 0$ and $g(1) = f(1) - 1 \leq 0$. Obviously $g(0) * g(1) \leq 0$ and applying Bolzano's theorem we're done. ■

Corollary 5.8. ***Bursuck Ullam's theorem***
If $S^1 \xrightarrow{f} \mathbb{R}$ then $\exists x \in S^1$ such that $f(x) = f(-x)$

Proof. **ni de conya begonya** ■

Theorem 5.9. *Let (X, τ_X) a topological space and $\{C_i\}_{i \in I}$ such that $C_i \subseteq X$ are connected $\forall i \in I$ and $\bigcap_{i \in I} C_i \neq \emptyset$.*
Then $\bigcup_{i \in I} C_i$ is connected.

Proof. Suppose that $\exists U, V \in \tau_X$ such that we have:

1. $(\bigcup_{i \in I} C_i) \subseteq U \cup V$.
2. $(\bigcup_{i \in I} C_i) \cap U \cap V = \emptyset$

Because $\bigcap_{i \in I} C_i = \emptyset$, $\exists p \in \bigcap_{i \in I} C_i$ and by the second property, $p \in U$ or $p \in V$. Supposing $p \in U$ we have $\boxed{\forall i \in I}$:

1. $C_i \cap U \neq \emptyset$ $(C_i \subseteq U \cup V)$
2. $C_i \cap U \cap V \subseteq (\bigcup_{i \in I} C_i) \cap U \cap V = \emptyset$
3. $C_i \subseteq (\bigcup_{i \in I} C_i) \subseteq U \cup V$

Now because C_i is a connected set $C_i \cap V = \emptyset \implies (\bigcup_{i \in I} C_i) \cap V = \emptyset$ and $\bigcup_{i \in I} C_i$ is a connected set. ■

Theorem 5.10. *(X, τ_X) a topological space and $A, B \subseteq X$ such that $A \subseteq B \subseteq \overline{A}$.*
Then A is a connected set $\implies$ B is a connected set.

Proof. If B is not a connected set, then $\exists U, V \in \tau_X$ such that:

1. $U \cap B \neq \emptyset \implies \exists p \in U \cap B$ and $p \in \overline{A}$ means $U \cap A \neq \emptyset$
2. $V \cap B \neq \emptyset \implies V \cap A \neq \emptyset$
3. $B \cap U \cap V = \emptyset \implies A \cap U \cap V = \emptyset$ $(A \subseteq B)$
4. $B \subseteq U \cup V \implies A \subseteq U \cup V$

Then A is not a connected set. ■

Theorem 5.11. *Let $\{(X_i, \tau_i)\}_{i \in I}$ be topological spaces.*
Then $X = \prod_{i \in I} X_i$ is connected $\iff$ $\forall i \in I$, X_i is a connected set (considering $\tau_X = \prod_{i \in I} \tau_i$.

Proof. $\boxed{\implies}$
Immediate because the projection is continuous and X is a connected set.
$\boxed{\impliedby}$

Finite case:
We are going to proof it by induction over n (then $X_1 \times \cdots \times X_N = (X_1 \times \ldots X_{N-1}) \times X_N)$).
Let $n = 2$. Fixing $b \in X_2$ we define $T = X_1 \times \{b\} \cong X_1$ and then T is connected. For $x \in X_1$ we define $T_x = \{x\} \times X_2 \cong X_2$ and then T_x is connected.
Because $T \cap T_x = \{x\} \times \{b\} \neq \emptyset$ then by 5.9 $T \cup T_x$ is a connected set.
Finally because $\bigcap_{x \in X_1} (T \cup T_x) = X_1 \times \{b\}$, $X_1 \times X_2 = \bigcup_{x \in X_1} (T \cup T_x)$ is connected by 5.9.

Infinite case:
Fix $a = (a_i)_{i \in I} \in \prod_{i \in I} X_i$ and for $J = \{j_1, \ldots, j_k\} \subset I$ we define $A_J = \prod_{j \in J} X_j \times \prod_{i \notin J} \{a_i\} \cong \prod_{j \in J} X_j$ and then A_J is connected by 5.9 and because J is finite.
Defining $A = \bigcup_{J \subset I \text{ and } J \text{ finite}} A_J$ and noting that $a \in \bigcap_{J \subset I \text{ and } J \text{ finite}} A_J \neq \emptyset$ then A is a connected set by 5.9. Obviously by 5.10 $\overline{A}$ is connected. Now we want to see that $\overline{A} = X$.
Then we want to see that if $U \in \tau_X$ then $U \cap A \neq \emptyset$. If $\mathcal{B}$ is a basis of X then $U = \bigcup_{\text{some } B \in \mathcal{B}} B$.
We are going to see that if $B \in \mathcal{B}$ then $B \cap A \neq \emptyset$.
B it's like $B = \prod_{j \in J} U_j \times \prod_{i \notin J} X_i$ for a J and $U_j \in \tau_j$. For that J we take $A_J = \prod_{j \in J} X_j \times \prod_{i \notin J} \{a_i\}$ note that $A_J \cap B = \prod_{j \in J} U_j \times \prod_{i \notin J} \{a_i\} \neq \emptyset$ and we're done because $A = \bigcup A_J$. ■

5.2 Arc-connected sets

Theorem 5.12. *$X \xrightarrow{f} Y$ a continuous function and X an arc-connected set. Then $f(X)$ is arc-connected.*

Proof. Taking $q_1, q_2 \in f(X)$ $(\exists p_1, p_2 \in X$ such that $f(p_1) = q_1, f(p_2) = q_2)$ and because X is arc-connected, $\exists \gamma : [0,1] \to X$ a continuous map such that $\gamma(0) = p_1$ and $\gamma(1) = p_2$.
Defining $\sigma = f \circ \gamma : [0,1] \to f(X)$ that is continuous because it's the composition of two continuous maps, we have $\sigma(0) = f(\gamma(0)) = q_1$ and $f(\gamma(1)) = q_2$. Then $f(X)$ is arc-connected. ■

Theorem 5.13. *Let (X, τ_X) a topological spae.*
Then if we have a collection $\{C_i\}_{i \in I}$ such that $C_i \subseteq X$, $\forall i, C_i$ are arc-connected and $\bigcap_{i \in I} C_i \neq \emptyset$. We have $\bigcup_{i \in I} C_i$ is arc-connected.

Proof. Taking $p \in \bigcap_{i\in I} C_i$ and $q_1, q_2 \in \bigcup_{i\in I} C_i$, $\exists i_1, i_2 \in I$ such that $q_1 \in C_{i_1}, q_2 \in C_{i_2}$. Because $p \in C_{i_1}$ and C_{i_1}, $q_1 \sim_{a.c} p$. By the same argument $q_2 \sim_{a.c} p$. By the transitivity of the equivalence relationship $q_1 \sim_{a.c} q_2$. ■

Theorem 5.14. *Let $\{(X_i, \tau_i)\}_{i\in I}$ a topological space. Then $X = \prod_{i\in I} X_i$ is arc-connected $\iff$ $\forall i \in I$ X_i is arc-connected.*

Proof. $\boxed{\implies}$

Direct

$\boxed{\impliedby}$

Take $p = (p_i)_{i\in I}, q = (q_i)_{i\in I} \in X$. Fix $i \in I$ and because X_i is arc-connected, $\exists \gamma_i : [0,1] \to X_i$ such that $\gamma_i(0) = p_i$ and $\gamma_i(1) = q_i$.
Define $\gamma : [0,1] \to X$ where $\gamma(t) = (\gamma_i(x_i))_{i\in I}$. γ is a continuous map because it is continuous component by component. Finally, $\gamma(0) = (p_i)_{i\in I} = p$ and $\gamma(1) = (q_i)_{i\in I} = q$ and X is arc-connected. ■

Theorem 5.15. *X arc-connected $\implies$ X connected*

Proof. Fixing $p \in X$, then $\forall q \in X$, $\exists \gamma_q : [0,1] \to X$, such that $\gamma_q(0) = p$, $\gamma(1) = q$, then because $\bigcap_{q\in X} \gamma_q([0,1]) = p \neq \emptyset$ we have that $X = \bigcup_{q\in X} \gamma_1([0,1])$ is a connected set because of 5.9 (note that $\gamma_q([0,1])$ is a connected set because it's the continuous image of a connected set). ■

Definition 5.2. If $[p]_c = \bigcup_{C \text{ connected}, p\in C} C$ and $[p]_{ac} = \bigcup_{A \text{ arc}-\text{connected}, p\in A} A$ then the connected components are the equivalence classes for $\sim_c$ and the arc-connected components are the equivalence classes for $\sim_{ac}$.

Theorem 5.16. *Let (X, τ_X) a topological space and:*

1. $X = \bigcup_{i\in I} C_i$
2. $X = \bigcup_{j\in J} A_j$

Then:

1. *The both items are partitions of X (meaning that if $i_1 \neq i_2$ then $C_{i_1} \cap C_{i_2} = \emptyset$)*
2. *If C is a connected set $\exists i$ such that $C \subseteq C_i$*
3. *If C is an arc-connected set $\exists j$ such that $A \subseteq A_j$*
4. *$\forall j, \exists i$ such that $A_j \subseteq C_i$*
5. *$\forall i, C_i$ is a closed set*

Proof.
1. Obviously because they are classes of equivalence and they cannot intersect.
2. Taking $p \in C$, $\exists i \in I$ such that $p \in C_i$, because C_i is the maximal connected set containing p, $C \subseteq C_i$
3. Analogue from 2

4. A_j is an arc-connected component $\implies$ A_j is an arc-connected $\implies$ A_j is a connected set $\implies$ $\exists i \in I$ such that $A_j \subseteq C_i$ (by 1)

5. C_i is a connected component $\implies$ C_i is a connected set $\implies$ $\overline{C_i}$ is a connected set and because $\overline{C_i} \cap C_i \neq \emptyset$ then $\overline{C_i} \subseteq C_i$. Finally, C_i is a closed set.

■

6 Aplications to the topology of $\mathbb{R}^2$

6.1 Definitions

Definition 6.1. (X, τ_X) is a contractil space $\iff$ $\exists p \in X$ such that $X \simeq \{p\}$

Definition 6.2. Let (X, τ_X) a topological space and $A \overset{i}{\hookrightarrow} X$ a subspace. We say that A is a deformation retract of X $\iff$ $\exists r : X \to A$ a continuous map such that:

1. $r \circ i = Id_A$
2. $i \circ r \simeq Id_X$

PD: Els de dins es queden a dins i els de fora van a un lloc que fa homotopia amb l'original

Definition 6.3. If $(X, \tau_X), (Y, \tau_Y)$ and $C^0(X, Y) = \{f : X \to Y | f \ cont\}$, then $[X, Y] = (C^0(X, Y)/\simeq)$ (set of equivalence classes of homotopy of continuous maps between X and Y) [dues classes son diferents si les funcions no son homòtopes entre elles]

Remark 6.1. if X is a contractil space and Y an arc-connected space then $[X, Y] = [\{p\}, Y]$ and it has $\boxed{1}$ element because Y is arc-connected, every two maps from X to Y can be continuously deformed to the other one.

6.2 Theorems

Theorem 6.2. *If $f, g : S^1 \to \mathbb{R}^2$ are continuous maps such that $p \notin f(S^1) \cup g(S^1)$ then*

$$\mu(f; p) = \mu(g; p) \iff f \simeq g \in \mathbb{R}^2 - \{p\}$$

Proof. $\boxed{\Longleftarrow}$
If $f \simeq g \in \mathbb{R}^2\{p\}$ then $N_p \circ f \simeq N_p \circ g \implies gr(N_p \circ f) = gr(N_p \circ g) \implies \mu(f; p) = \mu(g; p)$
$\boxed{\Longrightarrow}$

We have $\mu(f, p) = \mu(g, p) \implies gr(N_p \circ f) = gr(N_p \circ g) \implies N_p \circ f \simeq N_p \circ g$ (by the injectivity of the grade map). Define:

$$\begin{array}{rcl} i_p : S^1 & \longrightarrow & \mathbb{R}^2 - \{p\} \\ z & \longmapsto & p + z \end{array}$$

Then $N_p \circ f \simeq N_p \circ g \implies i_p \circ N_p \circ f \simeq i_p \circ N_p \circ g$
Now note that $i_p \circ N_p \simeq Id_{\mathbb{R}^2/p}$ because exists the homotopy:

$$\begin{array}{rcl} H : (\mathbb{R}^2 - \{p\}) \times I & \longrightarrow & \mathbb{R}^2 - \{p\} \\ (z, t) & \longmapsto & zt + (1-t)(p + \frac{z-p}{\|z-p\|}) \end{array}$$

Because it's well defined $f \simeq Id_{\mathbb{R}^2 - \{p\}} \circ f \simeq i_p \circ N_p \circ f \simeq i_p \circ N_p \circ g \simeq Id_{\mathbb{R}^2 - \{p\}} \circ g \simeq g$ ■

Corollary 6.3. ***Poincaré-Bohol theorem***
Let $f, g : S^1 \to \mathbb{R}^2$ be two continuous maps such that $p \notin f(S^1) \cup g(S^1)$.
If $\forall z \in S^1, \overline{f(z)g(z)} \not\ni p \implies \mu(f,p) = \mu(g,p)$

Proof. If $p \notin \overline{f(z)g(z)}$ then we can use the homotopy

$$\begin{aligned} H : (S^1) \times I &\longrightarrow \mathbb{R}^2 - \{p\} \\ (z,t) &\longmapsto f(z)t + (1-t)g(z) \end{aligned}$$

That's well defined because $p \notin f(S^1) \cup g(S^1)$, and it's continuous. Then $f \simeq g \implies \mu(f,p) = \mu(g,p)$. ■

Corollary 6.4. ***Rouché's theorem***
$f, g : S^1 \to \mathbb{R}^2$ a continuous map and $p \notin f(S^1) \cup g(S^1)$. Then if $\forall z \in S^1, d(f(z), g(z)) < d(g(z), p) \implies \mu(f,p) = \mu(g,p)$

Proof. $\forall z \in S^1, p \notin \overline{f(z)g(z)}$ (otherwise $d(f(z), g(z)) \geq d(g(z), p)$). Now we apply Poincaré Bohl's theorem. ■

Theorem 6.5. ***Bolzano's theorem (Dim 2)***
$F : \mathbb{D}^2 \to \mathbb{R}^2$ and $p \in \mathbb{R}^2$. Let $f = F_{|\partial \mathbb{D}^2}$.
Then if $\mu(f,p) \neq 0 \implies \exists a \in \mathbb{D}^2$ such that $f(a) = p$

Proof. Suppose $p \notin F(\mathbb{D}^2)$, then define:

$$\begin{aligned} g : (S^1) \times I &\longrightarrow \mathbb{D}^2 \\ (z,t) &\longmapsto zt \end{aligned}$$

and $H : S^1 \times I \to \mathbb{R}^2 - \{p\}$ such that $H = F \circ g$ and it's a continuous map. Then if $z \in S^1, H(z,0) = F(0)$ and $F(z,1) = f(z)$ and we are determining an homotopy between $F(0) \simeq f(z), z \in S^1$. Then $\mu(F(0), p) = \mu(f,p)$ what it's a contradiction because $\mu(F(0),p) = 0$ and $\mu(f,p) \neq 0$. ■

Theorem 6.6. ***Fundamental theorem of Algebra***
$p(z) \in \mathbb{C}[z], gr(p) = n \geq 1 \implies \exists a \in \mathbb{C}$ such that $p(a) = 0$

Proof. Suppose $p(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$ and define $M = 2 + |a_0| + \cdots + |a_{n-1}|$.
Define $l : S^1 \to \overline{B(0,M)}$ such that $l(z) = Mz$.
Now define: $f = p \circ l_{|S^1}, q(z) = z^n, g = q \circ l_{|S^1}$. Obviously $\mu(g,0) = gr(N_0 \circ g) = \cdots = n \neq 0$.
We want to apply Rouché's theorem, so we want to analize:

$$\begin{aligned} |f(z) - g(z)| &= |a_0 + a_1(Mz) + \cdots + a_{n-1}(Mz)^{n-1}| \\ &\leq |a_0| + |a_1|M + \cdots + |a_{n-1}|M^{n-1} \\ &< M^{n-1}(|a_0| + \cdots + |a_{n-1}|) \ \mid \ (M > 1 \implies M^{n-1} > M^i, i \in \{1, \ldots, n-2\}) \\ &= M^{n-1}(M-2) \\ &\leq M^n \\ &= |g(z) - 0|. \end{aligned}$$

By Rouché's theorem $\mu(f,0) = \mu(g,0) = n$. Apply Bolzan'os theorem and $\exists a \in S^1$ such that $(f \circ l)(a) = f(Ma) = 0, Ma \in \mathbb{C}$. ■

Lemma 6.7. *$\nexists r : \mathbb{D}^2 \to S^1$ continuous such that $r_{|S^1} = Id_{S^1}$*

Proof. Suppose that $\exists\ \mathbb{D}^2 \xrightarrow{r} S^1 \xhookrightarrow{i} \mathbb{R}^2$ continuous such that $r_{|S^1} = Id_{S^1}$. Define $f = i \circ r$ such that $f : \mathbb{D}^2 \to \mathbb{R}^2$.
Now $\mu(f_{|S^1}, 0) = gr(N_o \circ f_{|S^1}) = gr(\frac{z}{\|z\|}) = 1$ and by bolzano's theorem $\exists a \in \mathbb{D}^2$ such that $f(a) = 0$ what's a contradiction because $Im(r) \subseteq S^1$. ■

Theorem 6.8. ***Brower Fixed point theorem***
If $F : \mathbb{D}^2 \to \mathbb{D}^2$ is a continuous map $\implies$ $\exists x \in \mathbb{D}^2$ such that $F(x) = x$

Proof. Suppose $\forall x \in \mathbb{D}^2, F(x) \neq x$ and define: $\gamma(x) = F(x) + \lambda(x - F(x)), \lambda > 0$ be the map that sends the segment $\overline{xF(x)}$ to the first intersection with S^1. To have this conditions we should find λ such that $\|\gamma(x)\| = 1\ \forall x \in \mathbb{D}^2$.

$$\begin{aligned} 1 &= \|\gamma(x)\|^2 \\ &= \langle F(x) - \lambda(x - F(x)), F(x) - \lambda(x - F(x))\rangle \\ &= \|F(x)\|^2 + 2\lambda\langle F(x), x - F(x)\rangle + \lambda^2\|x - F(x)\|^2 \end{aligned}$$

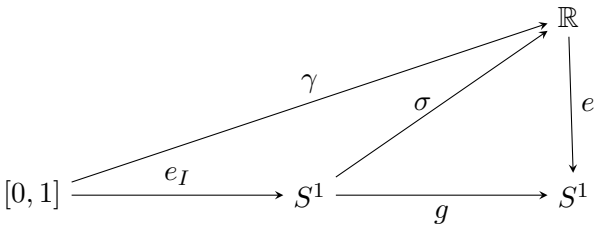
Hence,

$$\lambda = \frac{-2\langle F(x), x - F(x)\rangle + \sqrt{\dots}}{2\|x - F(x)\|^2}.$$

Then λ is continuous because it's a linear combination of operations with continuous maps. (inside the square root we have a positive thing). Finally $\gamma(x) : \mathbb{D}^2 \to S^1$ is a continuous map such that if $x \in S^1, \gamma(x) = F(x) + 1 \cdot (x - F(x)) = x$ and that's a contradiction because $\nexists$ a retraction with this properties by the last lemma. ■

Lemma 6.9. *Let $g : S^1 \to S^1$ a continuous map such that $\forall x \in S^1, g(-x) = -g(x)$. Then $gr(g) \neq 0$*

Proof. Let γ be an "aixecament" of g such that:



Suppose that $gr(g) = 0$, then $\gamma(1) = \gamma(0)$ and $\exists \sigma : S^1 \to \mathbb{R}$ continuous map such that $g = e \circ \sigma$. By Bursuck Ullam's theorem in Dim.1, $\exists x_0 \in S^1$ such that $\sigma(x_0) = \sigma(-x_0)$. Then $g(x_0) = e^{2\pi i\sigma(x_0)} = e^{2\pi i\sigma(-x_0)} = g(-x_0)$. By hypothesis $g(x_0) = 0$ what's a contradiction because $Im(g) \subseteq S^1$ ■

Theorem 6.10. ***Bursuck Ullam's theorem Dim.2***
If $F : S^2 \to \mathbb{R}^2$ continuous them $\exists p \in S^2$ such that $F(p) = F(-p)$

Proof. Define $G(x) = F(x) - F(-x)$, we want to find $x \in S^2$ such that $G(x) = 0$.
Define $l : \mathbb{D}^2 \to S^2$ like $l(x, y) = (x, y, \sqrt{1 - x^2 - y^2})$ and $\bar{G} = G \circ l : \mathbb{D}^2 \to \mathbb{R}^2$.

If $g = \bar{G}_{|S^1}$ we want to find that $\mu(g,0) \neq 0$ to apply Bolzano's theorem.
$\mu(g,0) = gr(N_o \circ g)$ and note that $N_o \circ g : S^1 \to S^1$ such that:

$$(N_o \circ g)(-x) = \frac{F(-x,-y,0) - F(x,y,0)}{\|F(-x,-y,0) - F(x,y,0)\|} = -\frac{F(x,y,0) - F(-x,-y,0)}{\|F(x,y,0) - F(-x,-y,0)\|} = -(N_o \circ g)(x)$$

We used that if $x \in S^1 \implies l(x)\vec{z} = 0$.
This map has the same characteristics as the one on the lemma, then $\mu(g,0) = gr(N_o \circ g) \neq 0$ and $\exists a \in \mathbb{D}^2$ such that $\bar{G}(a) = G(l(a)) = 0$, where $l(a) = x_0$. ■